GENERATIVE AI

Lab Experiment – 03:

1. **Implement a Generative Adversarial Network (GAN) to generate synthetic images or tabular data.**
   Dataset: MNIST (for images) or Synthetic Financial Transactions.

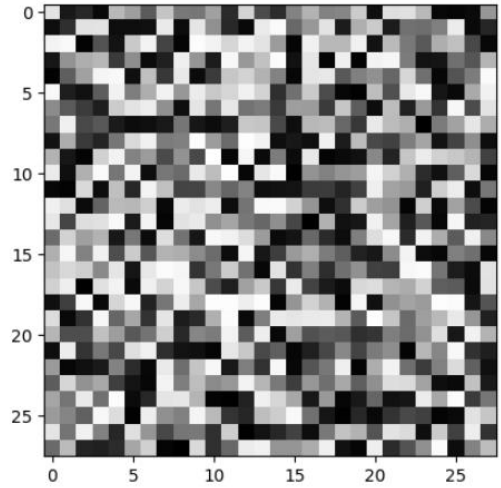
Code:

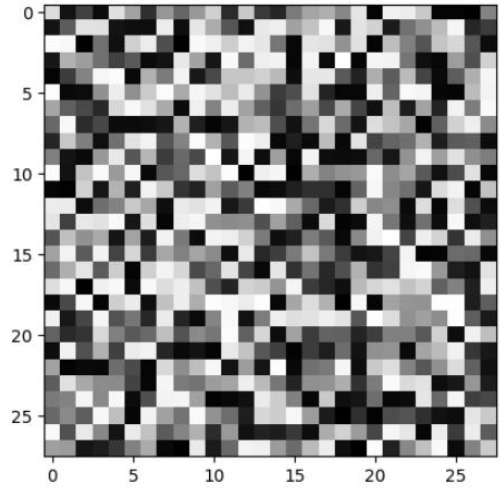
```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt


(x_train, _), (_, _) = tf.keras.datasets.mnist.load_data()
x_train = (x_train - 0.5) * 2
x_train = x_train.reshape(-1,28,28,1)

generator = tf.keras.Sequential([
    tf.keras.layers.Dense(128, activation='relu', input_dim=100),
    tf.keras.layers.Dense(784, activation='sigmoid'),
    tf.keras.layers.Reshape((28,28,1))
])

discriminator = tf.keras.Sequential([
    tf.keras.layers.Flatten(input_shape=(28,28,1)),
    tf.keras.layers.Dense(128, activation='relu'),
    tf.keras.layers.Dense(1, activation='tanh')
])
discriminator.compile(optimizer='adam', loss='binary_crossentropy')

discriminator.trainable = False
gan = tf.keras.Sequential([generator, discriminator])
gan.compile(optimizer='adam', loss='binary_crossentropy')

for epoch in range(2000):
    idx = np.random.randint(0, x_train.shape[0], 32)
    real_imgs = x_train[idx]

    noise = np.random.normal(0,1,(32,100))
    fake_imgs = generator.predict(noise, verbose=0)

    discriminator.train_on_batch(real_imgs, np.ones((32,1)))
    discriminator.train_on_batch(fake_imgs, np.zeros((32,1)))

    noise = np.random.normal(0,1,(32,100))
    gan.train_on_batch(noise, np.ones((32,1)))
```

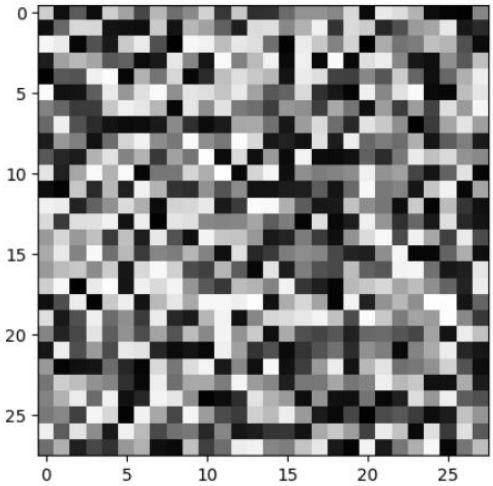
```
    if epoch % 100 == 0:
        print("Epoch:", epoch)
noise = np.random.normal(0,1,(5,100))
images = generator.predict(noise)

for i in range(5):
    plt.imshow(images[i].reshape(28,28), cmap='gray')
    plt.show()
```

OUTPUT:

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 29ms/step
```



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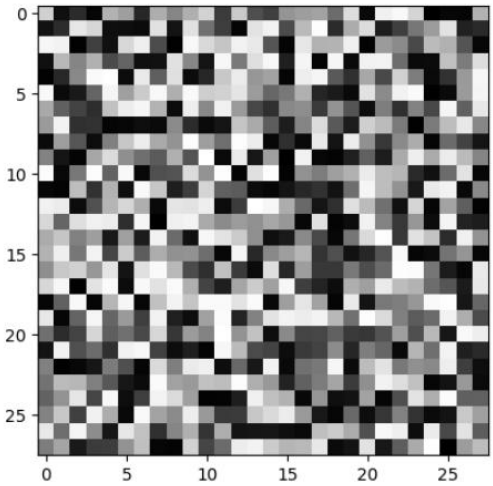
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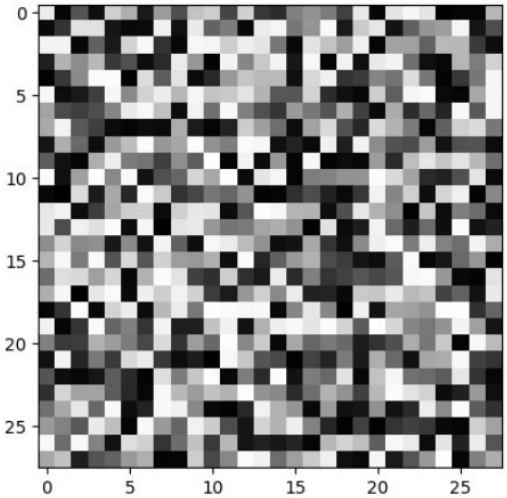
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